Specialization in Pyramidal Cell Structure in the Sensory-Motor Cortex of the Chacma Baboon (*Papio ursinus*) With Comparative Notes on Macaque and Vervet Monkeys

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ABSTRACT

The systematic study of pyramidal cell structure has revealed new insights into specialization of the phenotype in the primate cerebral cortex. Regional specialization in the neuronal phenotype may influence patterns of connectivity and the computational abilities of the circuits they compose. The comparative study of pyramidal cells in homologous cortical areas is beginning to yield data on the evolution and development of such specialized circuitry in the primate cerebral cortex. Recently, we have focused our efforts on sensory-motor cortex. Based on our intracellular injection methodology, we have demonstrated a progressive increase in the size of, the branching structure in, and the spine density of the basal dendritic trees of pyramidal cells through somatosensory areas 3b, 1, 2, 5, and 7 in the macaque and vervet monkeys. In addition, we have shown that pyramidal cells in premotor area 6 are larger, more branched, and more spinous than those in the primary motor cortex (MI or area 4) in the macaque monkey, vervet monkey, and baboon. Here we expand the basis for comparison by studying the basal dendritic trees of layer III pyramidal cells in these same sensory-motor areas in the chacma baboon. The baboon was selected because it has a larger cerebral cortex than either the macaque or vervet monkeys; motor cortex has expanded disproportionately in these three species; and motor cortex in the baboon reportedly has differentiated to include a new cortical area not present in either the macaque or vervet monkeys. We found, as in monkeys, a progressive increase in the morphological complexity of pyramidal cells through areas 3b, 5, and 7, as well as from area 4 to area 6, suggesting that areal specialization in microcircuitry was likely to be present in a common ancestor of primates. In addition, we found subtle differences in the extent of the interareal differences in pyramidal cell structure between homologous cortical areas in the three species.
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Key words: intracellular injection; Lucifer Yellow; dendritic spine; Sholl

Grant sponsor: the McDonnell Foundation; Grant sponsor: the Australian National Health and Medical Research Council; Grant sponsor: the Spanish Ministry of Science and Technology; Grant number: DGCYT PM99-0105 and BFI 2003-02745; Grant sponsor: the Comunidad Autonoma de Madrid; Grant number: 01/0782/2000; Grant sponsor: the South African National Research Foundation.

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Received 21 January 2005; Accepted 25 April 2005

DOI 10.1002/ar.a.20217

Published online 12 August 2005 in Wiley InterScience (www.interscience.wiley.com).

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Systematic differences in pyramidal cell structure have been demonstrated in cortical areas involved in somatosensory and motor processing in the macaque monkey, the vervet monkey, and human (Ramon y Cajal, 1899; Hendry and Jones, 1983; Lund et al., 1993; Elston and Rockland, 2002). Pyramidal cells in the primary somatosensory area are smaller, less branched, and less spinous than those in Brodmann's area 5, which are smaller, less branched, and less spinous than those in Brodmann's area 7. Those in primary motor cortex (Brodmann's area 4) are smaller, less branched, and less spinous than those in premotor cortex (Brodmann's area 6). Various mechanisms have been proposed as to how these differences in pyramidal cell structure may influence neural/cortical function in somatosensory and motor processing (for review, see Elston, 2002, 2003; Jacobs and Scheibel, 2002). For example, differences in the size of the dendritic arbors influence the size of the topographic representation from which they sample inputs. Differences in branching structure determine the potential for compartmentalization of processing within the dendritic tree. Differences in the number of spines reflect variation in the number of potential excitatory inputs to the arbors of individual neurons. Any of these structural differences potentially influence the biophysical properties of neurons and the memory capacity of the circuits they form (e.g., Poirazi and Mel, 2001; Stepanyants et al., 2002; Chklovskii et al., 2004).

Studies in visual cortex and granular prefrontal cortex in primates have revealed different trends in pyramidal cell specialization. For example, pyramidal cell structure is relatively constant in the primary visual area (V1) in the mature primate brain, despite considerable volumetric increases in the size of V1, whereas pyramidal cells have become notably more complex in structure during volumetric increase of inferotemporal cortex (Elston et al., 2005b). Pyramidal cells have undergone a remarkable degree of specialization during the evolution of granular prefrontal cortex (gPFC) in primates: those in human gPFC are more complex than those in macaque gPFC, which, in turn, are more complex than those in the gPFC of marmoset monkeys and galagos (Elston et al., 2001, 2005c). However, it remains unknown as to how the pyramidal cell phenotype may differ between homologous cortical areas in different primate species. Insufficient data have been published to allow meaningful speculation on how pyramidal cell circuitry in sensory-motor cortex may vary between species, and how that may parallel the evolution of bipedalism and forelimb dexterity in higher primates.

Here we studied pyramidal cell structure in the sensorymotor cortex of the Chacma baboon to better understand how the increasing size of the parietal and frontal lobes may have influenced the evolution of circuitry in sensorymotor cortex in various species, how circuitry in sensory motor cortex may differ between species, and how specialization in pyramidal cell structure/sensorimotor circuitry has evolved in different primate species. The baboon in particular was chosen for study for several reasons: it has a considerably larger brain then either the vervet or macaque monkeys; the relative size of its somatosensory and motor cortex is larger than that in either of the two monkey species; and the relative size of motor cortex in the baboon is larger than that in the macaque and vervet monkeys (Fig. 1), suggesting that it has undergone disproportionate expansion during its evolution (Brodmann, 1913). In addition, the baboon motor cortex reportedly contains a cortical area not identified in the macaque monkey (Watanabe-Sawaguchi et al., 1991).

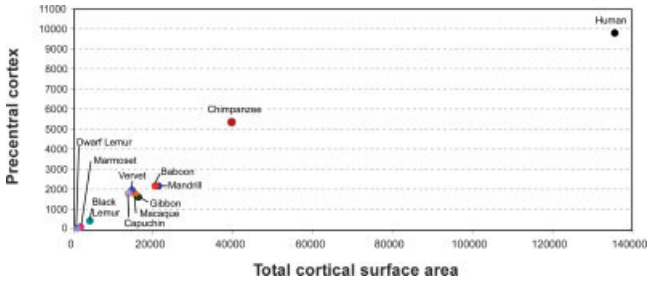


Fig. 1. Graph illustrating the relationship between the total cortical surface vs. the cortical surface area of motor cortex, defined as all nongranular cortex anterior to the central sulcus (Brodmann, 1913). Linear regression analysis revealed a close correlation between the two variables ($y = 0.0717x + 725.47$; $r^2 = 0.93$). Data on both axes are mm^2. No data are available for how many animals of each species were included in Brodmann's study.

We found, as in the macaque and vervet monkeys, that there were systematic differences in pyramidal cell structure between cortical areas involved in somatosensory and motor processing. More specifically, cells in premotor cortex are more complex than those in the primary motor area, and that cells became progressively more complex through somatosensory areas 3b, 5, and 7. We suggest that these specializations have taken place during the expansion of the parietal and frontal lobe and endow cortical circuitry in homologous cortical areas of these two species with different functional capabilities.

MATERIALS AND METHODS

Animals

Data were sampled from a wild-caught adult male Chacma baboon (*Papio ursinus*) weighing 23 kg. Cells were also injected in visual and cingulate cortex in this same animal (B2) (Elston et al., 2005a,d). Consistent with our other cell injection studies in primates, all data were sampled from the left hemisphere.

The exact age of the animal was unknown; however, several observations led us to believe that the animal was a sexually mature young adult. First, the animal was well developed physically, displaying impressive musculature, yet lacked serious injury or scarring usually observed in aged adults. Second, the animal had a full compliment of teeth, with relatively few cavities as compared to older animals. Third, there was no obvious sign of eye injury often seen in older animals.

The animal was anesthetized by intramuscular injection of a mixture of ketamine hydrochloride and xylazine. When deeply anesthetized, the animal was administered a lethal intravenous dose of sodium pentobarbital, then perfused through the heart with 5 L of 0.95% saline in 0.1 M phosphate buffer (PB), then 5 L 4% paraformaldehyde in PB. The brain was then removed from the animal and cortical blocks were dissected from the brain for cell injection.

Tissue Preparation

A block of tissue (1 cm wide) was taken along a rostrocaudal transect, approximately 5 mm from the midline,

extending from the arcuate sulcus to the parieto-occipital sulcus. White matter was trimmed to leave a block of relatively even thickness (approximately 4 mm thick). Separate blocks were then taken from the caudal bank of the central sulcus and the adjacent rostral bank of the central sulcus. The blocks were then prepared as flattened preparations by positioning each piece in a glass petri dish between a number of microscope slides. The number of slides was determined by the thickness of the block such that when a piece of glass was placed on top of the block, and gentle pressure was applied with weights, the lightly fixed tissue made contact on the upper and lower surface. The blocks were left overnight in 4% paraformaldehyde in PB. The next day, individual sections (250 μm) were cut tangential to the cortical surface of each block with the aid of a vibratome.

Intracellular Injection

Sections were prelabeled with 4,6-diamidino-2-phenylindole (DAPI; Sigma D9542), which fluoresces under UV excitation (see Fig. 2 of Elston and Rosa, 1997), enabling injection of individually labeled cell bodies under visual control (for details, see Buhl and Schlote, 1987; Buhl and Lübke, 1988; Einstein, 1988; Elston and Rosa, 1997). As seen in that figure, cortical layers are easily distinguished in DAPI-labeled sections, in both transverse and tangential sections. Based on the color of the sections, the density and distribution of neurons labeled with DAPI, and calculations of the relative depth of layer III from Nissl preparations (e.g., Fig. 2), we were able to select sections that contained specific layers for intracellular injection. Moreover, while injecting under visual control, it is easy to distinguish pyramidal cells from spiny nonpyramidal cells, allowing further confirmation of the cortical layers in which we injected cells. Only cells injected at the base of layer III are included here. Cells injected in other cortical layers will be published elsewhere.

Cells were injected with continuous current (up to 100 nA) until the individual dendrites of each cell could be traced to abrupt distal tips and the dendritic spines were easily visible. Cells were injected in the dorsolateral cortex medial to the superior precentral sulcus [area 6 of Brodmann (1907), corresponding to area 6aα of Watanabe-Sawaguchi et al. (1991) and area F2 of Geyer et al. (2000)], the rostral bank of the central sulcus [area 4 of Brodmann (1907) and Watanabe-Sawaguchi et al. (1991), corresponding to area F1 of Geyer et al. (2000)], the adjacent caudal bank of the central sulcus [area 3b of Vogt and Vogt (1919)], the exposed bank of the postcentral gyrus [Brodmann's area 5, corresponding to area PE of Geyer et al. (2000)], and the exposed bank of the medial portion of the intraparietal gyrus [area 7 of Vogt and Vogt (1919), corresponding to area PG of Geyer et al. (2000); Fig. 3]. By injecting cells along a rostrocaudal transect through these cortical areas, we increased the likelihood that we sampled data from similar body part representations in all areas, e.g., patterns of connections between these areas form rostrocaudal projections (Watanabe-Sawaguchi et al., 1991) as demonstrated in the macaque monkey (e.g., Jacobson and Trojanowski, 1977; Matsumura and Kubota, 1979; Barbas and Mesulam, 1981; Kawamura and Naito, 1984; Barbas and Pandya, 1987; Ghosh et al., 1987; Arikuni et al., 1988; Manger et al., 1997). Approximately 50 cells were injected in each cortical area.

Immunohistochemical Processing

Following cell injection, sections were processed with an antibody to Lucifer Yellow [1:400,000 in 2% bovine serum albumin (Sigma A3425), 1% Triton X-100 (BDH 30632), and 5% sucrose in PB] that was raised in rabbit (by staff at the Cajal Institute, Spain). The 250 μm sections were left in this solution for 5 days at room temperature to allow complete penetration throughout the depth of tissue. The sections were then washed three times in PB before being processed for 2 hr in biotinylated antirabbit (Amersham RPN 1004) at a concentration of 1:200 in 2% bovine serum albumin, 1% Triton X-100, and 5% sucrose in PB. Sections were then washed another three times in PB before being processed for a further 2 hr in streptavidin biotin horseradish peroxidase (Amersham RPN1051) at a concentration of 1:200 in PB. Sections were then washed another three times in PB, preincubated in DAB (3,3′-diaminobenzidine; Sigma D 8001), and then reacted with DAB/hydrogen peroxide (for details, see Elston and Rosa, 1997; Elston et al., 1997) to reveal a light-stable robust reaction product (Fig. 4).

Morphological Analyses

Neurons were drawn in two dimensions with the aid of a camera lucida coupled to a Zeiss Axioplan microscope (40× objective). Only spiny cells that had an unambiguous apical dendrite were included for analyses. Moreover, only cells that were completely filled, well impregnated, and whose entire basal dendritic arbor was contained within the section were included for analyses. Only dendrites that issued from the cell body were analyzed; apical collaterals were excluded as they were truncated.

The size of the basal dendritic arbour was determined with the aid of a digitizing tablet (Intuos, Wacom, Japan) and NIH image software (NIH Research Services Branch, Bethesda, MD). The areas were calculated as the interior of a polygon that joined the outermost distal tips of the basal dendrites. The branching structure was quantified by counting the number of dendritic intersections with concentric circles centered on the cell body (Sholl, 1953). Ten randomly selected horizontally projecting basal dendrites of different cells in each cortical area were drawn at high power (100× oil immersion objective) to allow quantification of dendritic spines. Spine density was determined by counting the number of spines per 10 μm segment along the entire length of each dendrite. Cell bodies were drawn at high power (100× oil immersion objective) and their size was determined by scanning the drawings and using features of NIH Image. All statistical analysis was performed with SPSS (Chicago, IL). All experiments were performed in accordance with guidelines endorsed by the NIH (publication 86-23, revised 1985) and approved by the University of the Witwatersrand Ethics Committee.

RESULTS

A total of 258 neurons were injected in layer III of cortical areas 3b, 4, 5, 6, and 7 in the left hemisphere of B2. One hundred and forty-one of these were included for analysis as they had an unambiguous apical dendrite characteristic of pyramidal neurons (for review, see Feldman, 1984; DeFelipe and Fariñas, 1992; Nieuwenhuys, 1994), had their complete basal dendritic arbors included within the section, and were well filled.

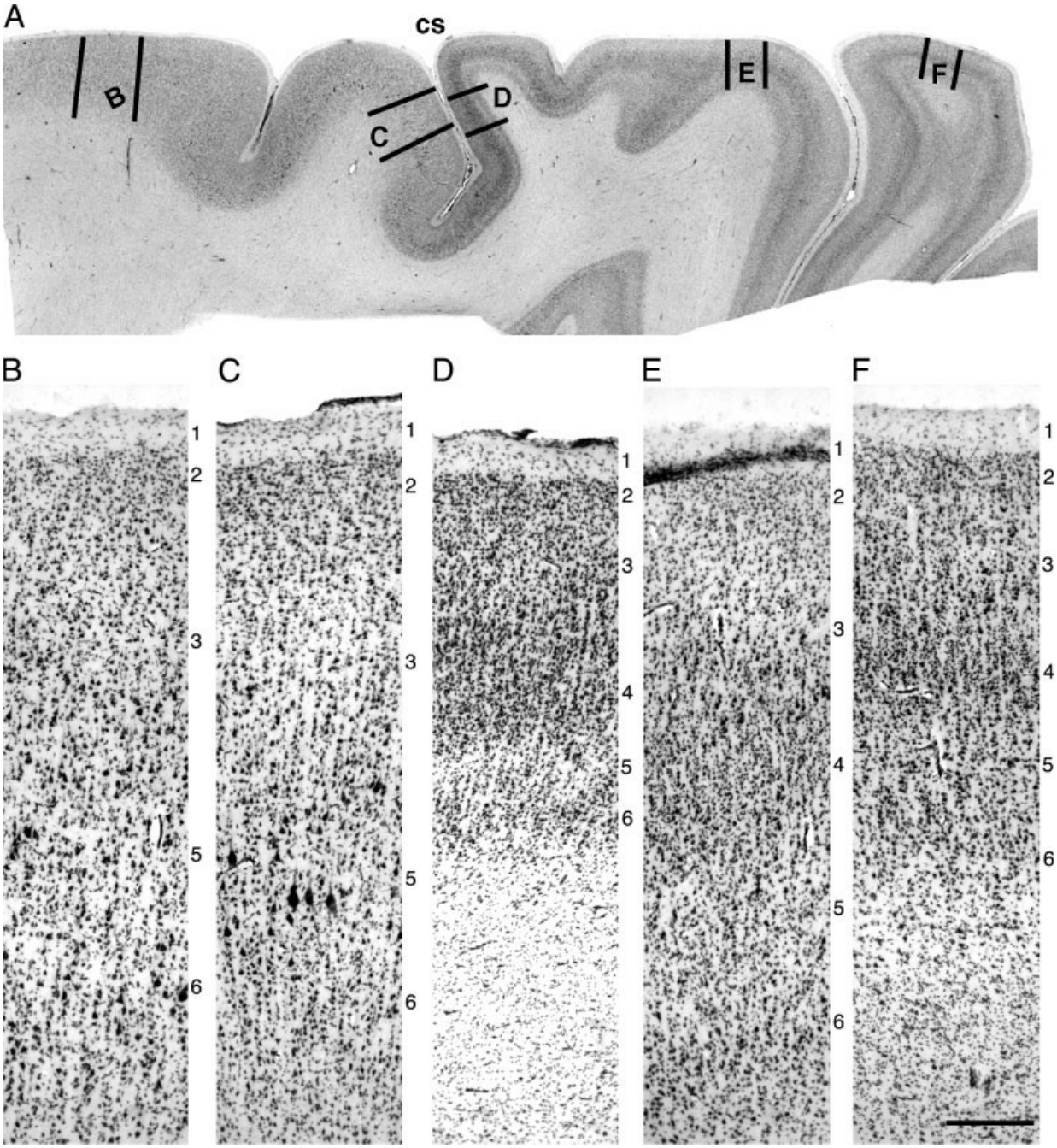


Fig. 2. Photomicrograph of (**A**) a single section take parallel to the midline encompassing the pre- and postcentral gyri from where neurons were sampled, which was processed for Nissl substance. Differences in cytoarchitecture make it possible to distinguish easily the border between the primary somatosensory area 3b and the primary motor area (area 4), as well as between area 4 and premotor cortex (area 6). In addition, it is possible to distinguish the border between area 3b and area 1. **B–F:** Higher-power photomicrographs of (B) area 6, (C) area 4, (D) area 3b, (E) area 5, and (F) area 7 illustrating differences in lamination patterns. CS, central sulcus. Scale bar = 250 μm in B–F.

Basal Dendritic Field Areas

The sizes of the basal dendritic arbors of layer III pyramidal cells differed between cortical areas (Fig. 5A, Table 1). Specifically, pyramidal cells in premotor cortex (Brodmann's area 6) had larger basal dendritic trees than those in the primary motor area (Brodmann's area 4). In addition, there was a systematic increase in the size of the dendritic trees of layer III pyramidal cells through somatosensory areas 3b, 5, and 7 (Fig. 5A, Table 1). An analysis of variance (ANOVA) revealed significant differences ($P < 0.05$) in the areas of the basal dendritic arbors of pyramidal cells in the different cortical areas ($F_{(4)} =$

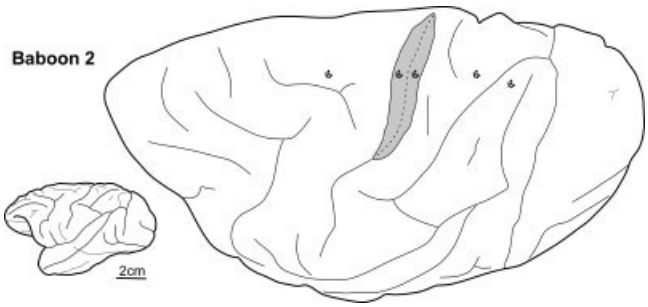


Fig. 3. Schematic showing the regions of cortex from which tissue was sampled from Chacma baboon 2 (B2).

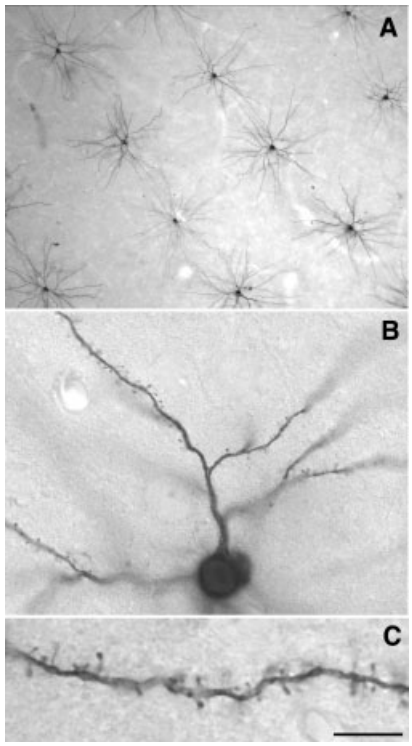


Fig. 4. Low- (**A**), intermediate- (**B**), and high- (**C**) power photomicrographs of layer III pyramidal cells injected with Lucifer Yellow and processed for a DAB (3,3′-diaminobenzidine) reaction product. Illustrated cells were sampled from cortical area 5. Scale bar = 400 μm in A, 20 μm in B, and 4 μm in C.

12.53). Posthoc pair-wise analyses revealed that the dendritic trees of cells in area 6 were significantly different ($P < 0.05$) to those in area 4 (B2: $F_{(1)} = 7.62$). Differences in the size of the dendritic trees of cells in somatosensory areas 3b and 5 were not significant ($F_{(1)} = 3.09$; $P = 0.09$). The dendritic trees of cells in area 7 were significantly larger than those in area 5 ($F_{(1)} = 42.17$). Moreover, the dendritic trees of cells in area 4 were significantly larger ($P < 0.05$) than those in area 3b (B2: $F_{(1)} = 5.68$).

Branching Patterns of Basal Dendritic Arbors

In order to quantify differences in branching structure, we performed Sholl analysis. From Figure 5B, it can be seen that cells in area 4 had fewer branches that those in area 6 (Table 2). In addition, cells in area 3b had, on average, fewer branches than those in area 5, which, in turn, had fewer dendritic branches than those in area 7. Moreover, the peak branching complexity (the maximum average number of dendritic intersections per circle) in area 4 was less than that in area 6, and that in area 3b was less than that in area 5, which, in turn, was less than that in area 7 (Table 2).

In order to determine whether these differences were statistically significant, we performed repeated-measures ANOVAs (cortical area by distance from the soma, 5×12 design), which revealed a P value of less than 0.001 ($F_{(1,4)} = 17.25$). Posthoc repeated-measures ANOVAs revealed that cells in area 3b were significantly ($P < 0.05$) less branched than those in area 5 (B2: $F_{(1,1)} = 7.45$; $P < 0.05$) and that cells in area 5 were significantly less branched than those in area 7 ($F_{(1,1)} = 29.81$). In addition, cells in area 4 were significantly less branched than those in area 6 ($F_{(1,1)} = 16.16$).

Spine Densities of Basal Dendrites

Visual observation revealed differences in the density of spines on the basal dendrites of pyramidal neurons in the different cortical areas. In order to quantify the differences, spines were counted per 10 μm dendritic segment from the cell body to the distal tips of the dendrites (Fig. 5C, Table 3). More than 7,000 individual spines were reconstructed from 50 different dendrites (10 per cortical area, 5 areas). Several trends emerge from these plots. First, as reported elsewhere, spine density was 0 in the proximal 10 μm of the basal dendrites, rose to a peak density at approximately one-third the distance from the soma to the distal tips of the dendrites, and decreased with progression toward the distal tips. Second, there was a trend for increased spine density from area 4 to 6 and another from area 3b to 5 to 7. Repeated-measures ANOVA (5×26 design) revealed a significant difference ($P < 0.001$) in spine densities between cortical areas ($F_{(1,4)} = 6.13$). Posthoc analyses revealed that spine density along the dendrites of cells in area 3b was significantly less ($P < 0.05$) than that in area 5 ($F_{(1,1)} = 7.67$). There was no significant difference in spine density along the basal dendrites of pyramidal cells in areas 5 and 7 ($F_{(1,1)} = 4.39$), nor between areas 4 and 6 ($F_{(1,1)} = 0.007$; $P = 0.94$).

In order to study the effects of differing branching patterns and spine densities of neurons in the different cortical areas, we made estimates of the total number of dendritic spines in the basal dendritic tree of the "average" layer III pyramidal neurone in each area. This was done by summing the product of the average spine density by the average number of dendritic branches over successive annuli of increasing radii (25 μm) (Elston, 2001). These calculations revealed that the average neurone in area 3b had fewer spines than that in area 5, which, in turn, had fewer spines than that in area 7 (1,963, 2,731, 4,685, respectively). In addition, the average neurone in area 4 had fewer spines than that in area 6 (3,510 and 4,552, respectively). Furthermore, our estimates revealed that the average neurone in area 3b had fewer spines than that in area 4.

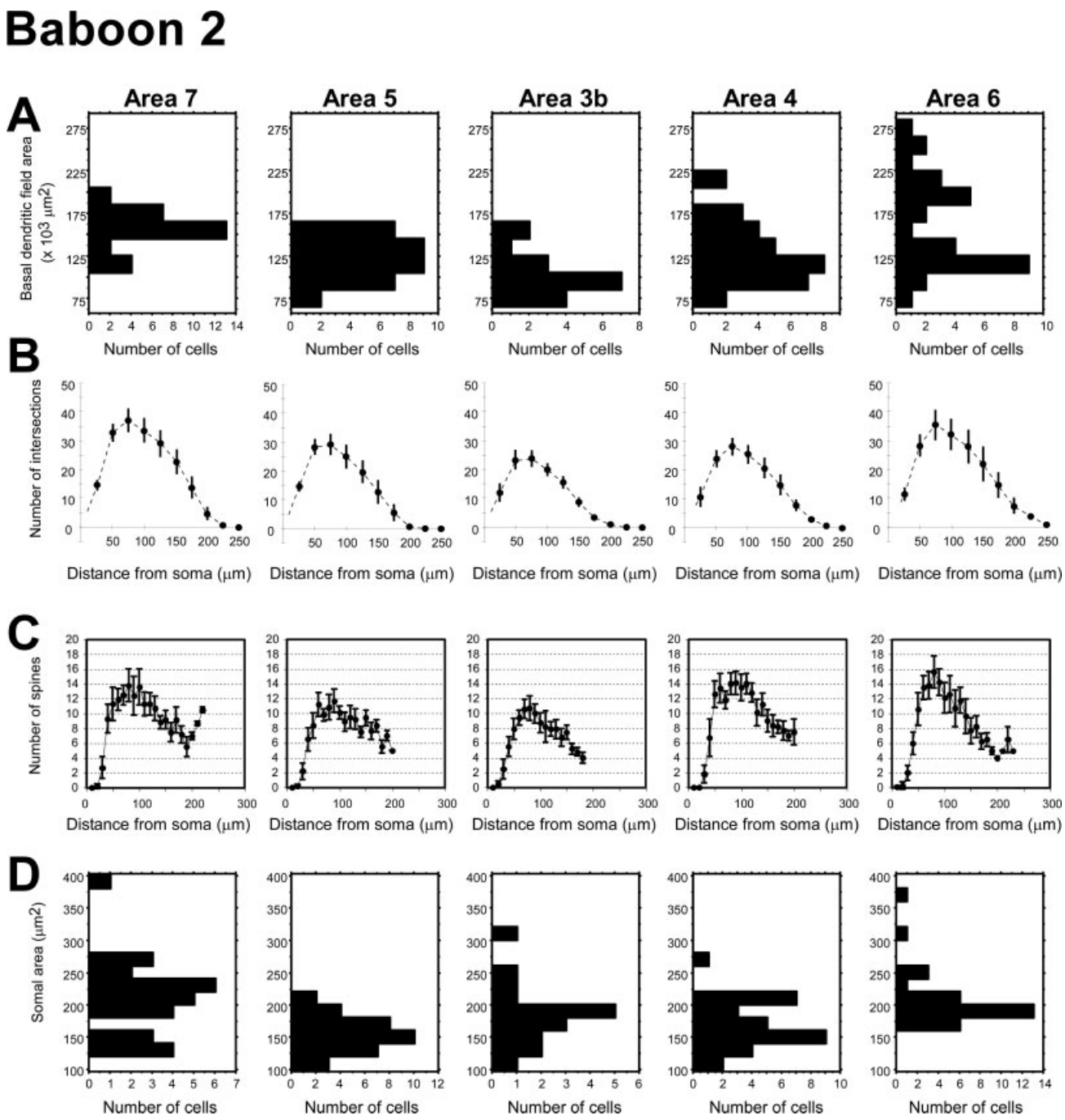


Fig. 5. Morphological data for cells in baboon 2. **A:** Frequency histograms of the size of the basal dendritic arbors of layer III pyramidal neurons in somatosensory areas 3b, 5, and 7 and motor areas 4 and 6. **B:** Graphs of the results of Sholl analysis of the basal dendritic trees of pyramidal neurons in sensory-motor areas 3b, 5, 7, 4, and 6. **C:** Plots of the spine density (per 10 μm increment) as a function of distance from the cell body to the distal tips of the dendrites of randomly selected basal dendrites of layer III pyramidal neurons in of areas 3b, 5, 7, 4, and 6. **D:** Frequency histograms of the size of the cell bodies (in the tangential plane) of pyramidal neurons in areas 3b, 5, 7, 4, and 6. Error bars = standard errors.

Somal Areas

Somal areas were drawn in the plane tangential to the cortical layers and plotted in Figure 5D (see also Table 4). Interestingly, the systematic trends reported for the size of the basal dendritic trees of layer III pyramidal cells were not necessarily paralleled in the size of their cell bodies. Cell bodies in area 3b, for example, were on average larger than those in both areas 4 and 5. Statistical analyses revealed significant differences ($P < 0.001$) in the size of the cell bodies of neurons between the five cortical

TABLE 1. Basal dendritic field area (μm^2) of layer III pyramidal cells in sensory-motor cortex of the Chacma baboon

Cortical area	n	Mean, $\mu m^2 \times 10^4$	SD $\times 10^4$	SEM $\times 10^4$	Minimum $\mu m^2 \times 10^4$	Maximum $\mu m^2 \times 10^4$
6	31	15.99	5.39	0.97	7.97	28.03
4	31	12.77	3.62	0.65	7.55	21.43
3b	17	10.43	2.40	0.58	6.70	14.93
5	34	12.22	2.28	0.39	8.22	16.09
7	28	15.62	2.29	0.43	10.81	20.45

areas ($F_{(4)} = 9.25$). Posthoc repeated-measures ANOVAs revealed that somata in area 3b were significantly ($P < 0.05$) different to those in area 5 ($F_{(1,1)} = 6.52$), and those of cells in area 5 were significantly different to those in area 7 ($F_{(1)} = 21.36$). In addition, the somata of cells in area 4 were significantly different to those in area 6 ($F_{(1)} = 14.69$). There was also a significant difference in the size of somata of cells in areas 3b and 4 ($F_{(1)} = 7.52$).

DISCUSSION

The present investigation is the third in a series of ongoing studies into species specialization in pyramidal cell structure in sensory-motor cortex of primates. These investigations are primarily aimed at correlating pyramidal cell structure to volumetric differences of the cortical areas, with the hope of better understanding the evolution of the cerebral cortex in primates (for review, see Elston, 2005f). Previously, we documented variation in pyramidal cell structure in different cortical areas involved in somatosensory and motor processing in monkeys belonging to the family Cercopithicidae. More specifically, we quantified the structure of the basal dendritic trees of layer III pyramidal cells in somatosensory areas 1, 2, 3b, 5, and 7 and motor areas 4 and 6 in the macaque and vervet (Savannah guenon) monkeys (Elston and Rockland, 2002). We found that there was a general trend for more spinous cells with posterior progression through 3b, 1, 2, 5, and 7 and anterior progression from area 4 to area 6. Here we quantified the structure of the basal dendritic trees of layer III pyramidal cells in homologous cortical areas in the Chacma baboon. We found, as in these monkeys, a trend for more spinous cells with progression through somatosensory cortical areas 3b, 5, and 7 and motor areas 4 and 6 (Fig. 6). In conjunction, these results reveal several things in relation to specialization in pyramidal cell structure during the evolution of sensory-motor cortex in primates: it was likely that regional specialization in the pyramidal cell phenotype was present in a common ancestor of extant primates; the consistent trend for a systematic increase in the number of spines in the dendritic trees of cells with progression through areas 3b, 5, and 7, as well as through 4 and 6, is likely to be important for specialist functions performed by neurons in these cortical areas; and relative species variation in the structure of pyramidal cells in homologous cortical areas are likely to impart different computational abilities in their respective circuitry. Each of these points is discussed further below.

Phylogenetic Relationships of Baboon, Macaque Monkey, and Vervet Monkey

Although there are varying reports, the lineage leading to present-day baboons is widely believed to have separated from the lineage leading to the vervet monkey some 10 million years ago (Fig. 7). That leading to present-day macaque monkeys separated from that leading to baboons approximately 7 million years ago. During this same period, chimpanzees and gorillas have evolved. During a shorter period (approximately 6 million years), various hominids have evolved and become extinct (e.g., *H. habilis*, *H. erectus*, *H. habilis*, *A. africanis*, *A. afarensis*, *A. biosei*, *A. robustus*). Clearly, enough time has passed for aspects of the baboon brain to have evolved its own unique specialized structure and functions different from those in the macaque and vervet monkeys.

Functional Implications of Morphological Specialization in Sensory-Motor Cortex

In the mature cerebral cortex, each dendritic spine receives at least one asymmetrical synapse (Colonnier, 1968; Jones, 1968), the presynaptic terminal of which contains glutamate (DeFelipe et al., 1988; Kharazia and Weinberg, 1993, 1999; Kharazia et al., 1996). Thus, differences in the number of spines in the dendritic trees of pyramidal cells reported here in the different sensory-motor cortical areas suggest that they receive different numbers of excitatory inputs (Fig. 6). In addition, differences in the branching patterns in dendritic trees reportedly influence the degree to which processing may be compartmentalized within the arbors (Koch et al., 1982): cells with many branches such as those in premotor cortex and area 7 have greater representational power than those with fewer branches such as those in MI and 3b (Poirazi and Mel, 2001).

As the pyramidal cell forms and receives most excitatory projections within a cortical area, both horizontal (McGuire et al., 1991; Melchitzky et al., 1998; Tanigawa et al., 1998) and vertical (White, 1989; DeFelipe and Fariñas, 1992), regional variations in the pyramidal cell phenotype are likely to reflect fundamentally different patterns of connectivity: cortex composed of larger, more spinous pyramidal cells is likely to be characterized by more complex patterns of connectivity than that composed of small, less spinous cells (for review, see DeFelipe et al., 2002; Elston, 2003). Based on their modeling studies, Stepanyants et al. (2002) have suggested that differences in these aspects of neuron structure allow different potential for plastic change, being greater in cells with more complex structure than those with less complex structure. Modeling studies also predict that the memory capacity of cortical circuits is influenced by the morphological complexity of the dendritic trees of the component neurons. Circuits composed of neurones of more complex structure have greater memory capacity than those composed of neurons of less complex structure (for review, see Chklovskii et al., 2004).

TABLE 2. Results of Sholl analysis of the basal dendritic arbors of layer III pyramidal cells sampled in baboon 2 (mean ± SEM)

B2	25 μm	50 μm	75 μm	100 μm	125 μm	150 μm	175 μm	200 μm	225 μm	250 μm	275 μm	300 μm
6	11.39 ± 0.35	28.19 ± 0.69	35.26[a] ± 1.21	31.90 ± 1.76	27.81 ± 1.85	21.74 ± 1.94	14.52 ± 2.06	7.19 ± 1.48	3.58 ± 0.90	0.87 ± 0.33	0.23 ± 0.15	0.13 ± 0.10
4	10.55 ± 0.44	23.74 ± 1.10	27.87 ± 0.96	28.45[a] ± 0.98	20.48 ± 0.99	14.61 ± 1.15	7.65 ± 1.13	2.74 ± 0.064	0.84 ± 0.34	0.19 ± 0.12		
3b	12.06 ± 0.59	23.47 ± 1.34	23.88[a] ± 1.55	20.00 ± 1.25	15.59 ± 0.96	8.71 ± 0.87	3.29 ± 0.65	0.76 ± 0.33	0.18 ± 0.13	0.06 ± 0.06		
5	14.49 ± 0.42	28.18 ± 0.76	28.97[a] ± 1.12	24.69 ± 1.23	19.44 ± 1.13	12.54 ± 1.19	5.28 ± 0.82	0.67 ± 0.18	0.15 ± 0.13			
7	14.52 ± 0.56	32.76 ± 1.02	37.00[a] ± 1.42	33.41 ± 1.46	28.97 ± 1.59	22.52 ± 1.49	13.62 ± 1.32	4.52 ± 0.73	0.62 ± 0.19	0.14 ± 0.011	0.10 ± 0.01	0.03 ± 0.03

[a]Peak average dendritic branching complexity. The number of cells sampled in each cortical area (n) is the same as for Table 1.

While these theories remain controversial, there exists a parallel between the complexity of the dendritic trees of pyramidal cells in different cortical regions and their discharge properties. For example, pyramidal cells in V1 are characterized by phasic discharge and have relatively simple structure (Ashford and Fuster, 1985; Elston and Rosa, 1997, 1998), those in inferotemporal and parietal association cortex are characterized by tonic discharge properties and are of intermediate complexity (Fuster and Jervey, 1982; Miller et al., 1993; Nakamura and Kubota, 1995; Constantinidis and Steinmetz, 1996; Elston et al., 1999a; Elston and Rosa, 2000), and those in granular prefrontal cortex are characterized by mnemonic processing (Fuster et al., 1982; Funahashi et al., 1989, 1993) and have among the most complex structure of all pyramidal cells in the cerebral cortex (Elston et al., 2001; Jacobs et al., 2002). Direct evidence that cortical circuitry composed of pyramidal cells of different structural complexity is characterized by different biophysical properties was reported by Murayama et al. (1997). These authors demonstrated that 40–100 Hz stimulation to layers 2/3 results in long-term potentiation in temporal lobe cortex but long-term depression in V1. How then do the differences in pyramidal cell structure reported here relate to somatosensory and motor processing?

As somatosensory and motor cortex is topographically organized (Kaas, 1990; Kaas and Garraghty, 1991; Pons et al., 1992), differences in the size of the dendritic trees of neurones in these different cortical areas will influence the proportion of the body surface they receive inputs from or the muscle groups they ultimately drive. For example, cells in area 3b, which have relatively small dendritic trees, sample inputs over a smaller proportion of the body map than cells in area 5, which, in turn, sample inputs from a smaller portion of the body map than cells in area 7. In motor cortex, cells in area 4 sample inputs from less extensive groups of muscles than those in area 6. These differing sampling geometries may be important for punctate vs. global somatosensation and precision motor control vs. locomotion.

In addition, the data from somatosensory and motor cortex (Elston and Rockland, 2002; Elston et al., 2005f; present results) provide examples of both a confirmation and contradiction to the trend for neurons of increasingly more complex structure reported in the proposed visual hierarchies (Elston et al., 1999a, 1999b). For example, there is a progressive increase in the size, branching complexity, and the total number of spines in the dendritic trees of neurons with progression through areas 3b, 1, 2, 5, and 7b, which are reportedly successive hierarchical levels in somatosensory cortex (Friedman, 1983; Friedman et al., 1986; Felleman and Van Essen, 1991: but see Neal et al., 1987; Andersen et al., 1990). In motor cortex, however, cells in area 4, which is generally considered to be the end of a cortical motor decision-execution pathway (Jones, 1986), are smaller, less branched, and less spinous than those in area 6 (Elston and Rockland, 2002; Elston et al., 2005f; present results; see also Jacobs et al., 2001). Moreover, cells in area 6 are smaller, less branched, and less spinous than those in granular prefrontal cortex (Elston, 2000; Elston et al., 2001), where some purposive movements are thought to originate (Paulesu et al., 1997; Geyer et al., 2000; Calvert, 2001; Friston, 2001; Elston, 2005). Clearly, more remains to be determined regarding the implications of regional variation in cell structure and

TABLE 3. Spine density per 10 μm of horizontally projecting dendrites as a function of distance from the cell body to the distal tips of the dendrites of layer III pyramidal neurons sampled in areas 3b, 4, 5, 6, and 7 of the Chacma baboon*

B2	0–10 μm	11–20 μm	21–30 μm	31–40 μm	41–50 μm	51–60 μm	61–70 μm	71–80 μm	81–90 μm	91–100 μm
6	0.10 ± 0.10	0.30 ± 0.30	2.00 ± 0.63	4.00 ± 0.95	10.60 ± 1.38	13.50 ± 1.01	13.60 ± 1.16	15.60 ± 1.36	14.20 ± 1.13	12.10 ± 1.29
4	0	0	1.80 ± 0.77	6.70 ± 1.59	12.60 ± 1.14	13.40 ± 1.27	11.80 ± 0.77	14.00 ± 0.94	14.10 ± 0.99	13.50 ± 1.13
3b	0	0.50 ± 0.27	2.50 ± 0.86	5.50 ± 0.83	7.90 ± 0.91	9.40 ± 0.64	10.60 ± 0.81	10.70 ± 1.02	10.00 ± 0.80	8.70 ± 0.83
5	0	0.20 ± 1.33	2.20 ± 0.69	6.50 ± 0.98	8.40 ± 1.07	11.20 ± 0.01	9.80 ± 0.53	10.80 ± 1.07	11.70 ± 0.98	10.10 ± 0.55
7	0	0.20 ± 0.20	2.70 ± 0.93	9.30 ± 1.20	11.30 ± 1.41	11.90 ± 0.92	12.50 ± 0.83	13.50 ± 1.39	12.40 ± 1.63	13.63 ± 1.54

B2	101–110 μm	111–120 μm	121–130 μm	131–140 μm	141–150 μm	151–160 μm	161–170 μm	171–180 μm	181–190 μm	191–200 μm
6	12.60 ± 1.58	10.70 ± 1.56	10.60 ± 1.57	8.70 ± 1.69	7.10 ± 1.35	7.40 ± 1.19	5.00 ± 1.02	3.90 ± 1.15	1.50 ± 0.78	0.80 ± 0.53
4	14.00 ± 0.84	12.80 ± 0.81	10.10 ± 1.12	10.10 ± 1.44	8.10 ± 1.29	7.30 ± 1.21	4.90 ± 1.39	2.30 ± 1.23	1.40 ± 0.95	1.50 ± 1.07
3b	8.20 ± 1.34	7.90 ± 0.59	7.10 ± 1.10	5.40 ± 1.13	5.20 ± 1.25	2.60 ± 0.92	2.40 ± 0.83	0.80 ± 0.55	–	–
5	8.90 ± 0.81	9.50 ± 1.00	9.20 ± 0.87	7.50 ± 0.45	9.40 ± 0.67	6.30 ± 1.20	5.00 ± 1.41	2.20 ± 0.95	1.40 ± 0.95	0.50 ± 0.50
7	11.30 ± 1.00	11.30 ± 0.97	10.70 ± 1.04	8.80 ± 0.55	9.20 ± 0.79	7.50 ± 0.79	6.40 ± 1.67	4.30 ± 1.29	2.20 ± 1.02	2.10 ± 1.08

B2	201–210 μm	211–220 μm	221–230 μm	231–240 μm	241–250 μm	251–260 μm	261–270 μm	271–280 μm	281–290 μm	291–300 μm
6	1.00 ± 0.67	1.30 ± 0.94	0.50 ± 0.50	–	–	–	–	–	1	–
4	–	–	–	–	–	–	–	–	–	–
3b	–	–	–	–	–	–	–	–	–	–
5	–	–	–	–	–	–	–	–	–	–
7	2.60 ± 1.33	2.10 ± 1.40	–	–	–	–	–	–	–	–

*Data are averaged over 10 randomly selected dendrites of different neurons in each cortical area (mean ± SEM).

TABLE 4. Somal size of layer III pyramidal cells in sensory-motor cortex of the Chacma baboon

Cortical area	n (μm^2)	Mean (μm^2)	SD	SEM	Minimum (μm^2)	Maximum (μm^2)
6	31	207.25	42.04	7.55	161.65	361.47
4	31	169.84	34.45	6.19	113.78	263.73
3b	17	183.00	46.14	11.19	108.74	300.29
5	34	156.94	26.85	4.61	105.27	206.79
7	28	207.35	56.37	10.65	122.02	382.19

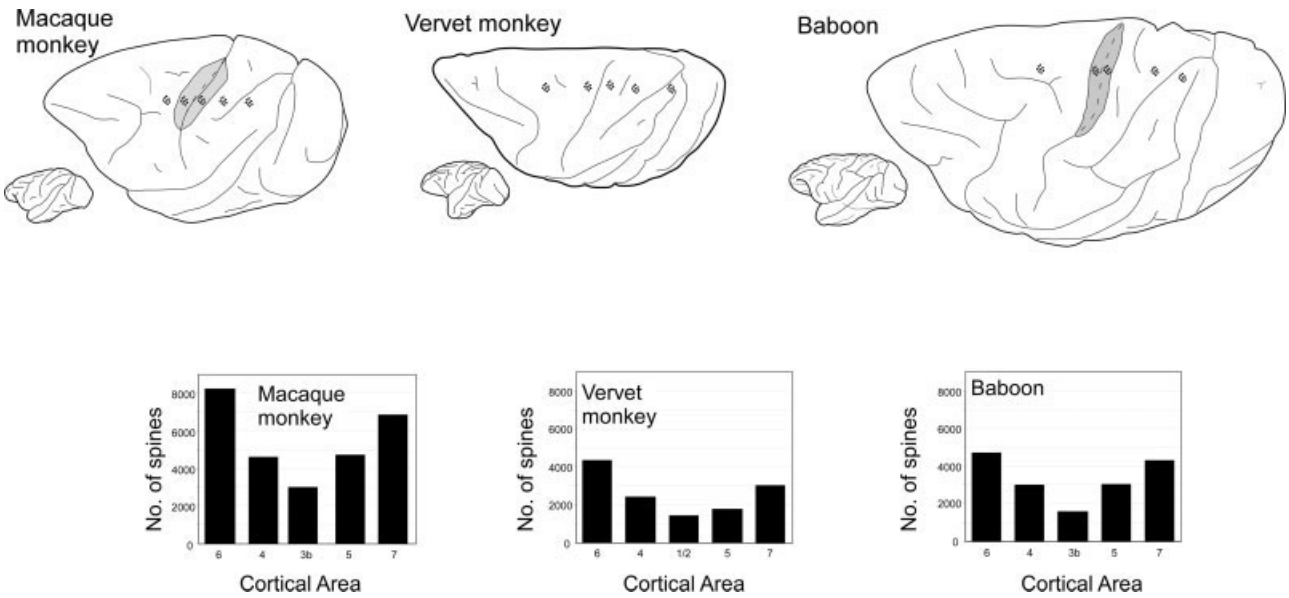


Fig. 6. Bar graphs illustrating our estimates of the total number of spines in the basal dendritic tree of layer III pyramidal cells in somatosensory (3b, 5, and 7) and motor (4 and 6) cortex in the baboon, vervet monkey, and macaque monkey. Note the consistent trends observed in the different species but the relative differences in the absolute values.

processing within sensory-motor cortex, particularly in view of the results of imaging studies, which suggest distributed cortical processing (for review, see Kaas, 1990; Mountcastle, 1995).

Caveats

Some of the comparative data were sampled from animals of unknown age. In addition, animals were reared

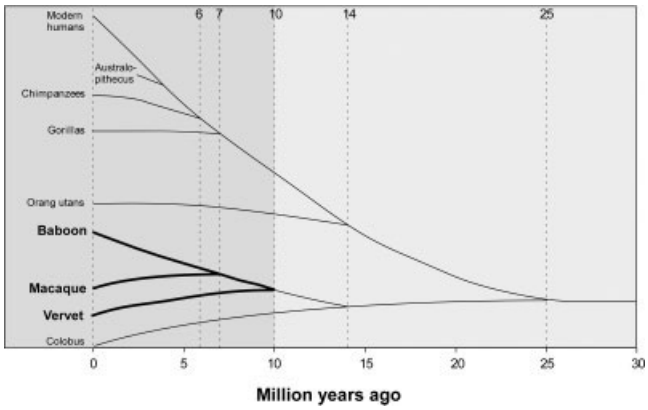


Fig. 7. Schematic illustrating primate phylogeny. Of particular interest are the estimates for which the lineages leading to present-day baboons, macaque monkey, and vervet monkeys diverged from each other, and which other primates species have evolved during that period. Modified from Dawkins, 2004.

under different conditions (macaque and vervet monkeys raised in colonies, baboons raised in the wild). Both aging and rearing conditions have been demonstrated to influence neuron structure (Volkmar and Greenough, 1972; Greenough et al., 1973; Scheibel et al., 1975; Nakamura et al., 1985; Jacobs and Scheibel, 1993; Anderson and Rutledge, 1996; Jacobs et al., 1997). However, data sampled from the cingulate cortex of these same animals (MF1, MF2, VM1, VM2, B1, B2) reveal similar numbers of spines in the dendritic trees of pyramidal cells in the macaque monkey, vervet monkey, and baboon (Elston et al., 2005a,e,g). It becomes increasingly less likely that species specializations in the pyramidal cell structure can be solely attributed to sampling error when different trends are found consistently in different cortical regions (e.g., visual vs. prefrontal). Instead, it is becoming more likely that there is phenotypic variation in pyramidal cell structure even between homologous cortical areas in primate species (for review, see Elston, 2005). It remains to be determined to what extent differences in the body part representations from which data were sampled may have influenced the interspecies differences in sensory-motor cortex. For example, data sampled from the baboon and vervet monkey were likely sampled from the lower limb representation, whereas those in the macaque monkey were likely sampled from the upper limb/hand representation (Powell and Mountcastle, 1959; Nelson et al., 1980; Pons et al., 1985, 1987; Waters et al., 1990).

The systematic intracellular injection of large numbers of pyramidal cells in functionally related cortical areas is beginning to reveal marked variation in their phenotypes. In the present series of investigations of pyramidal cell structure in the sensory-motor cortex in the baboon, vervet monkey, and macaque monkey, we have revealed systematic differences in pyramidal cell structure across species. These data suggest that regional specialization in pyramidal cell structure was a feature present in sensory-motor cortex of a common ancestor of extant primates. These findings are contrary to the belief that the cerebral cortex is uniform in structure and computational abilities. Instead, our data suggest that patterns of connectivity in, and the computational power of, cortical circuitry may vary between cortical regions and species.

LITERATURE CITED

Andersen RA, Asanuma C, Essick G, Siegel RM. 1990. Corticocortical connections of anatomically and physiologically defined subdivisions within the inferior parietal lobule. J Comp Neurol 296:65–113.

Anderson B, Rutledge V. 1996. Age and hemisphere effects on dendrite structure. Brain 119:1983–1990.

Arikuni T, Watanabe K, Kubota K. 1988. Connections of area 8 with area 6 in the macaque monkey. J Comp Neurol 277:21–40.

Ashford JW, Fuster JM. 1985. Occipital and inferotemporal responses to visual signals in the monkey. Exp Neurol 90:444–446.

Barbas H, Mesulam M-M. 1981. Organization of afferent input to subdivisions of area 8 in the rhesus monkey. J Comp Neurol 200: 407–431.

Barbas H, Pandya DN. 1987. Architecture and frontal cortical connections of the premotor cortex (area 6) in the rhesus monkey. J Comp Neurol 256:211–228.

Brodmann K. 1907. Beiträge zur histologischen Lokalisation der Großhirnrinde: die Cortexgliederung des Menschen. J Psychol Neurol 10:1–16.

Brodmann K. 1913. Neue forschungsergebnisse der grosshirnrindenanatomie mit besonderer berücksichtung anthropologischer fragen. Geselesch Deuts Naturf Artze 85:200–240.

Buhl EH, Schlote W. 1987. Intracellular Lucifer Yellow staining and electronmicroscopy of neurons in slices of fixed epitumourous human cortical tissue. Acta Neuropathol 75:140–146.

Buhl EH, Lübke J. 1988. Intracellular Lucifer Yellow injection in fixed brain slices combined with retrograde tracing, light and electron microscopy. Neuroscience 28:3–16.

Calvert GA. 2001. Crossmodal processing in the human brain: insights from functional neuroimaging studies. Cereb Cortex 11: 1110–1123.

Chklovskii DB, Mel BW, Svoboda K. 2004. Cortical rewiring and information storage. Nature 431:782–788.

Colonnier M. 1968. Synaptic patterns on different cell types in the different laminae of the cat visual cortex. Brain Res 9:268–287.

Constantinidis C, Steinmetz MA. 1996. Neuronal activity in posterior parietal area 7a during the delay periods of a spatial memory task. J Neurophysiol 76:1352–1355.

Dawkins R. 2004. The ancestor's tale. A pilgramage to the dawn of life. Weidenfeld & Nicolson.

DeFelipe J, Conti F, Van Eyck SL, Manzoni T. 1988. Demonstration of glutamate-positive axon terminals forming asymmetric synapses in cat neocortex. Brain Res 455:162–165.

DeFelipe J, Fariñas I. 1992. The pyramidal neuron of the cerebral cortex: morphological and chemical characteristics of the synaptic inputs. Prog Neurobiol 39:563–607.

DeFelipe J, Alonso-Nanclares L, Arellano JI. 2002. Microstructure of the neocortex: comparative aspects. J Neurocytol 31:299–316.

Einstein G. 1988. Intracellular injection of Lucifer Yellow into cortical neurons in lightly fixed sections and its application to human autopsy material. J Neurosci Meth 26:95–103.

Elston GN, Rosa MGP. 1997. The occipitoparietal pathway of the macaque monkey: comparison of pyramidal cell morphology in layer III of functionally related cortical visual areas. Cereb Cortex 7:432–452.

Elston GN, Pow DV, Calford MB. 1997. Neuronal composition and morphology in layer IV of two vibrissal barrel subfields of rat cortex. Cereb Cortex 7:422–431.

Elston GN, Rosa MGP. 1998. Morphological variation of layer III pyramidal neurones in the occipitotemporal pathway of the macaque monkey visual cortex. Cereb Cortex 8:278–294.

Elston GN, Tweedale R, Rosa MGP. 1999a. Cortical integration in the visual system of the macaque monkey: large scale morphological differences of pyramidal neurones in the occipital, parietal and temporal lobes. Proc R Soc Lond Ser B 266:1367–1374.

Elston GN, Tweedale R, Rosa MGP. 1999b. Cellular heterogeneity in cerebral cortex. A study of the morphology of pyramidal neurones in visual areas of the marmoset monkey. J Comp Neurol 415:33–51.

Elston GN. 2000. Pyramidal cells of the frontal lobe: all the more spinous to think with. J Neurosci RC20:1–4.

Elson GN, Rosa MGP. 2000. Pyramidal cells, patches, and cortical columns: a comparative study of infragranular neurons in TEO, TE, and the superior temporal polysensory area of the macaque monkey. J Neurosci 20:RC117 (111–115).

Elston GN. 2001. Interlaminar differences in the pyramidal cell phenotype in cortical areas 7m and STP (the superior temporal polysensory area) of the macaque monkey. Exp Brain Res 138:141–152.

Elston GN, Benavides-Piccione R, DeFelipe J. 2001. The pyramidal cell in cognition: a comparative study in human and monkey. J Neurosci RC21:1–5.

Elston GN. 2002. Cortical heterogeneity: implications for visual processing and polysensory integration. J Neurocytol 31:317–335.

Elston GN, Rockland K. 2002. The pyramidal cell of the sensorimotor cortex of the macaque monkey: phenotypic variation. Cereb Cortex 10:1071–1078.

Elston GN. 2003. Cortex, cognition and the cell: new insights into the pyramidal neuron and prefrontal function. Cereb Cortex 13:1124–1138.

Elston GN. 2005. The pyramidal cell in primate evolution. In: Kaas JH, Preuss TM, editors. Evolution of nervous systems. Oxford: Elsevier.

Elston GN, Benavides-Piccione R, Elston A, Manger P, DeFelipe J. 2005a. Regional specialization in pyramidal cell structure in the limbic cortex of the vervet monkey (*Cercopithecus pygerythrus*): an intracellular injection study of the anterior and posterior cingulate gyrus. Exp Brain Res (in press).

Elston GN, Elston A, Kaas JH, Casagrande V. 2005b. Regional specialization in pyramidal cell structure in the visual cortex of the galago: an intracellular injection study of striate and extrastriate areas with comparative notes on New World and Old World monkeys. Brain Behav Evol 66:10–21.

Elston GN, Elston A, Casagrande VA, Kaas J. 2005c. Pyramidal neurons of granular prefrontal cortex the galago: complexity in the evolution of the psychic cell. Anat Rec 285A:610–618.

Elston GN, Benavides-Piccione R, Elston A, DeFelipe J, Manger P. 2005d. Pyramidal cell specialization in the occipitotemporal cortex of the Chacma baboon (*Papio ursinus*). Exp Brain Res (in press).

Elston GN, Benavides-Piccione R, DeFelipe J. 2005e. A study of pyramidal cell structure in the cingulate cortex of the macaque monkey with comparative notes on inferotemporal and primary visual cortex. Cereb Cortex 15:64–73.

Elston GN, Benavides-Piccione R, Elston A, DeFelipe J, Manger P. 2005f. Specialization in pyramidal cell structure in the sensorymotor cortex of the vervet monkey (*Cercopithecus pygerythrus*). Neuroscience (in press).

Elston GN, Benavides-Piccione R, Elston A, DeFelipe J, Manger P. 2005g. Specialization in pyramidal cell structure in the cingulate cortex of the Chacma baboon (*Papio ursinus*): an intracellular injection study of the posterior and anterior cingulate gyrus. Brain Behav Evol (in press).

Feldman D. 1984. Morphology of the neocortical pyramidal neuron. In: Peters A, Jones EG, editors. Cerebral cortex, vol. 1, cellular components of the cerebral cortex. New York: Plenum. p 123–200.

Felleman DJ, Van Essen DC. 1991. Distributed hierarchical processing in primate cerebral cortex. Cereb Cortex 1:1–47.

Friedman DP. 1983. Laminar patterns of termination of corticocortical afferents in the somatosensory system. Brain Res 273:147–151.

Friedman DP, Murray EA, O'Neill JB, Mishkin M. 1986. Cortical connections of the somatosensory fields of the lateral sulcus of macaques: evidence for a corticolimbic pathway for touch. J Comp Neurol 252:323–347.

Friston KJ. 2001. Brain function, nonlinear coupling, and neuronal transients. Neuroscientist 7:406–418.

Funahashi S, Bruce CJ, Goldman-Rakic PS. 1989. Mnemonic coding of visual space in the monkey's dorsolateral prefrontal cortex. J Neurophysiol 61:331–349.

Funahashi S, Chafee MV, Goldman-Rakic PS. 1993. Prefrontal activity in rhesus monkeys performing a delayed anti-saccad task. Nature 365:753–756.

Fuster JM, Jervey JP. 1982. Neuronal firing in the inferotemporal cortex of the monkey in a visual memory task. J Neurosci 2:361–375.

Fuster JM, Bauer RH, Jervey JP. 1982. Cellular discharge in the dorsolateral prefrontal cortex of the monkey in cognitive tasks. Exp Neurol 77:679–694.

Geyer S, Matelli M, Luppino G, Zilles K. 2000. Functional neuroanatomy of the primate isocortical motor system. Anat Embryol 202:443–474.

Ghosh KK, Brinkman K, Porter R. 1987. A quantitative study of the distribution of neurons projecting to the precentral motor cortex in the macaque monkey (*M. fascicularis*). J Comp Neurol 259:424–444.

Greenough W, Volkmar FR, Juraska JM. 1973. Effects of rearing complexity on dendritic branching in frontolateral and temporal cortex in the rat. Exp Neurol 41:371–378.

Hendry SHC, Jones EG. 1983. The organization of pyramidal and non-pyramidal cell dendrites in relation to thalamic afferent terminations in the monkey somatic sensory cortex. J Neurocytol 12:277–298.

Jacobs B, Scheibel AB. 1993. A quantitative analyses of Wernicke's area in humans: I, lifespan changes. J Comp Neurol 327:83–96.

Jacobs B, Larsen-Driscoll L, Schall M. 1997. Lifespan dendritic and spine changes in areas 10 and 18 of human cortex: a quantitative Golgi study. J Comp Neurol 386:661–680.

Jacobs B, Schall M, Prather M, Kapler L, Driscoll L, Baca S, Jacobs J, Ford K, Wainwright M, Treml M. 2001. Regional dendritic and spine variation in human cerebral cortex: a quantitative study. Cereb Cortex 11:558–571.

Jacobs B, Scheibel AB. 2002. Regional dendritic variation in primate cortical pyramidal cells. In: Schüz A, Miller R, editors. Cortical areas: unity and diversity. London: Taylor and Francis. p 111–131.

Jacobson S, Trojanowski J. 1977. Prefrontal granular cortex of the rhesus monkey: I, interhemispheric cortical afferents. Brain Res 132:209–233.

Jones EG. 1968. An electron microscopic study of the terminations of afferent fiber systems onto the somatic sensory cortex of the cat. J Anat 103:595–597.

Jones EG. 1986. Connectivity of the primate sensory-motor cortex. In: Peters A, Jones EG, editors. Cerebral cortex, vol. 5, sensory motor areas and aspects of cortical connectivity. New York: Plenum. p 113–182.

Kaas JH. 1990. The somatosensory system. In: Paxinos G, editor. The human nervous system. San Diego, CA: Academic Press. p 813–844.

Kaas JH, Garraghty PE. 1991. Hierarchical, parallel, and serial arrangements of sensory cortical areas: connection patterns and functional aspects. Curr Opin Neurobiol 1:248–251.

Kawamura K, Naito J. 1984. Corticocortical projections to prefrontal cortex in the rhesus monkey investigated with horseradish peroxidase techniques. Neurosci Res 1:89–103.

Kharazia VN, Weinberg RJ. 1993. Glutamate in terminals of the thalamocortical fibers in rat somatic sensory cortex. Neurosci Lett 157:162–166.

Kharazia VN, Phend KD, Rustioni A, Weinberg RJ. 1996. EM colocalization of AMPA and NMDA receptor subunits at synapses in rat cerebral cortex. Neurosci Lett 210:37–40.

Kharazia VN, Weinberg RJ. 1999. Immunogold localization of AMPA and NMDA receptors in somatic sensory cortex of albino rat. J Comp Neurol 412:292–302.

Koch C, Poggio T, Torre V. 1982. Retinal ganglion cells: a functional interpretation of dendritic morphology. Phil Trans R Soc Lond B 298:227–264.

Lund JS, Yoshioka T, Levitt JB. 1993. Comparison of intrinsic connectivity in different areas of macaque monkey cerebral cortex. Cereb Cortex 3:148–162.

Manger PR, Woods A, Munoz A, Jones EG. 1997. Hand/face border as a limiting boundary in the body representation in monkey somatosensory cortex. J Neurosci 17:6338–6351.

Matsumura M, Kubota K. 1979. Cortical projection to hand-arm motor area from posterior arcuate areas in macaque monkey: a histological study of retrograde transport of horseradish peroxidase. Neurosci Lett 11:241–246.

McGuire BA, Gilbert CD, Rivlin PK, Wiesel TN. 1991. Targets of horizontal connections in macaque primary visual cortex. J Comp Neurol 305:370–392.

Melchitzky DS, Sesack SR, Pucak ML, Lewis DA. 1998. Synaptic targets of pyramidal neurons providing intrinsic horizontal connections in monkey prefrontal cortex. J Comp Neurol 390:211–224.

Miller EK, Li L, Desimone R. 1993. Activity of neurons in anterior inferior temporal cortex during a short-term memory task. J Neurosci 13:1460–1478.

Mountcastle VB. 1995. The evolution of ideas concerning the function of neocortex. Cereb Cortex 5:289–295.

Murayama Y, Fujita I, Kato M. 1997. Contrasting forms of synaptic plasticity in monkey inferotemporal and primary visual cortices. Neuroreport 8:1503–1508.

Nakamura S, Akiguchi I, Kameyama M, Mizuno N. 1985. Age-related changes of pyramidal cell basal dendrites in layers III and V of human motor cortex: a quantitative Golgi study. Acta Neuropathol 65:281–284.

Nakamura K, Kubota K. 1995. Mnemonic firing of neurons in the monkey temporal pole during a visual recognition memory task. J Neurophysiol 74:162–178.

Neal JW, Pearson RCA, Powell TPS. 1987. The corticocortical connections of area 7b, PF in the parietal lobe of the monkey. Brain Res 419:341–346.

Nelson RJ, Sur M, Felleman DJ, Kaas JH. 1980. Representations of the body surface in the postcentral parietal cortex of *Macaca fasicularis*. J Comp Neurol 192:611–643.

Nieuwenhuys R. 1994. The neocortex: an overview of its evolutionary development, structural organization and synaptology. Anat Embrol 190:307337.

Paulesu E, Frackowiak RSJ, Bottini G. 1997. Maps of somatosensory systems. In: Frackowiak RSJ, Friston KJ, Frith CD, Dolan RJ, Mazziotta JC, editors. Human brain function. London: Academic Press. p 183–242.

Poirazi P, Mel B. 2001. Impact of active dendrites and structural plasticity on the storage capacity of neural tissue. Neuron 29:779–796.

Pons TP, Garraghty PE, Cusick CG, Kaas JH. 1985. The somatotopic organization of area 2 in macaque monkeys. J Comp Neurol 241: 445–466.

Pons TP, Garraghty PE, Friedman DP, Mishkin M. 1987. Physiological evidence for serial processing in somatosensory cortex. Science 237:417–420.

Pons TP, Garraghty PE, Mishkin M. 1992. Serial and parallel processing of tactual information in somatosensory cortex of rhesus monkeys. J Neurophysiol 68:518–527.

Powell TPS, Mountcastle VB. 1959. Some aspects of the functional organization of the cortex of the postcentral gyrus of the monkey: a correlation of findings obtained in single unit analysis with cytoarchitecture. Bull Johns Hopkins Hosptl 105:133–162.

Ramon y Cajal S. 1899. Estudios sobre la corteza cerebral humana II: estructura de la corteza motriz del hombre y mamíferos superiores. Rev trim Micrográf 4:117–200.

Scheibel ME, Lindsay RD, Tomiyasu U, Scheibel AB. 1975. Progressive dendritic changes in the ageing human cortex. Exp Neurol 47:392–403.

Sholl DA. 1953. Dendritic organization in the neurons of the visual and motor cortices of the cat. J Anat 87:387–406.

Stepanyants A, Hof PR, Chklovskii DB. 2002. Geometry and structural plasticity of synaptic connectivity. Neuron 34:275–288.

Tanigawa H, Fujita I, Kato M, Ojima H. 1998. Distribution, morphology, and γ-aminobutyric acid immunoreactivity of horizontally projecting neurons in the macaque inferior temporal cortex. J Comp Neurol 401:129–143.

Vogt C, Vogt O. 1919. Allgemeine ergebnisse unserer hirnforschung. J Psychol Neurol 25:279–462.

Volkmar FR, Greenough W. 1972. Rearing complexity affects branching compexity in the visual cortex of the rat. Science 176:1445–1447.

Watanabe-Sawaguchi K, Kubota K, Arakuni T. 1991. Cytoarchitecture and intrafrontal connections of the frontal cortex of the brain of the hamadryas baboon (*Papio hamandryas*). J Comp Neurol 311: 108–133.

Waters RS, Samulack DD, Dykes RW, McKinley PA. 1990. Topographic organization of baboon primary motor cortex: face, hand, forelimb, and shoulder representaion. Som Mot Res 4:485–514.

White EL. 1989. Cortical circuits: synaptic organization of the cerebral cortex: structure, function and theory. Boston: Birkhaüser.